

Greenhouse gas emissions from silicon production - development of carbon footprint with changing energy systems

G. Saevarsdottir¹, H. Kvande², T. Magnusson³

¹Reykjavik University, Iceland, gudrunsa@ru.is

²Formerly NTNU, Norway, halvor.kvande@gmail.com

³Normi, Iceland, thordmag1@gmail.com

Keywords: Silicon production, greenhouse gas emissions, energy mix, carbon footprint, process emissions

Abstract – As the world pushes for a low-carbon future, populous regions of the world develop rapidly, increasing the global demand for materials. Emissions from the primary production of metals can be split into three parts; a) direct process related emissions, b) emissions due to auxiliary processes and c) indirect emissions due the production of the electrical energy used in the production process. As production of metallurgical grade (MG) silicon is energy intensive, the indirect emissions from the electrical energy production contribute a significant part of the overall specific carbon footprint per kg of silicon produced. In this paper, the trend in the energy use for MG silicon production on a global basis from 1995 is tracked, and the development of the carbon footprint from the energy mix is estimated. Silicon production increased by 240% from 2000 to 2019, and by 456% from 1995. As most of this increase in production has occurred in Asia, and more specifically China, the energy mix for the production site determines the indirect carbon dioxide emissions. The ratio of low carbon electric power sources such as hydro power or nuclear energy is still significant in the energy mix for silicon production. It was around 57% in 2019 as compared to 67% in 2000 and 71% in 1995, while the share of coal thermal power has increased from 20% in 1995 to 40% in 2019. This increased share of coal-based power has led to an increase in the indirect energy related carbon emissions to 6.3 kg CO₂e/kg Si in 2019, up from 3.3 in 1995, and is now similar to the direct process emissions that are around 4 kg CO₂e/kg Si. That is indeed a development in the wrong direction for the silicon industry, facing increasing climate related pressures.

INTRODUCTION

The demand, production and use of Metallurgical Grade Silicon metal (MG-Si) have been steadily increasing in recent decades, and the annual global production has increased by a factor of 2.4 since 2000. The emissions of greenhouse gases have also increased from this production, not only the process related emissions increasing linearly with the production volume, but also due to the limited supply of electrical energy generated from low carbon energy resources such as renewable sources and nuclear energy, leading to increased use of electric power generated from fossil fuel. In 2019, 43% of the MG-silicon production was based on electricity from fossil fuels, mainly coal (40%) and natural gas (3%).

In the most recent report from IPCC (IPCC 2018) it is stated that in order to prevent 1.5°C of global temperature increase, the total greenhouse gas emissions must fall by 45% from 2010 levels by 2030, and reach “net zero” around 2050. The silicon industry has to contribute to reach this ambitious goal. They can do many things, as will be discussed here, it is very important to de-carbonize the electrical energy production for the production process by maximizing the use of emission free power sources and minimizing the use of low-grade coal.

This can be obtained by increasing the silicon production in regions where CO₂-emission free electricity is available, or changing the energy mix for the production.

GREENHOUSE GAS EMISSION SOURCES FROM MG-SILICON PRODUCTION

From the quartz and coal mines to the cast MG-silicon, the total global average emissions are now between 10.2 and 12.6 metric tonnes of CO₂-equivalents per tonne of MG-silicon produced (t CO₂e/t Si), based on both direct process and indirect emissions (mainly from energy sources), depending on the various estimates and assumptions made in the literature. In any case these emissions are high, and the industry will surely face pressure to reduce them in the near future.

The three main contributions to this total value are (Monsen 1998, Kero 2016):

- The carbothermic reduction of silicon from quartz in the submerged arc furnace is around 4.7 – 5 t CO₂e/t Si, dependent on literature source, including contribution from the electrodes. This number also depends on the share of biocarbon reductants.
- The remainder is *indirect* emissions from MG-silicon production, with a global average of about 5.5 t CO₂e/t Si in 2019. The source of these emissions comes from generation of the electrical energy used to provide heat for the reactions in the submerged arc furnace. The specific electric energy consumption for MG-silicon production is in the range 10.5 – 12 kWh/kg Si. In addition, around 1 kWh/kg Si should be added to account for auxiliaries, such as operation of baghouse, fans, etc. Globally this category spans from practically zero when using nuclear or renewable power sources such as hydro, geothermal, wind or solar, and up to 15-21 t CO₂e/t Si, when coal is used for power production, dependent on energy source and the actual specific energy consumption (Bruckner 2014,). The emission estimates in this paper are based on source emission estimates published by IPCC (Bruckner 2014) and are towards the lower end of this estimate, while using the Ecoinvent database would give higher emission number (World Aluminum 2017).
- The mining and transport of raw materials, including electrode materials, are not included in the current estimate but these are low compared to the other factors.

As discussed in the next section, the opportunities to reduce the direct process emissions from the carbothermic silicon production process using existing technology are limited, and the paper will therefore focus on the development in the indirect emissions from power production.

EMISSIONS FROM THE SILICON METAL PRODUCTION PROCESS

Silicon is produced by carbothermic reduction in submerged arc furnaces. The raw materials for the process are SiO₂ in the form of quartz and carbon materials that are a mixture of coal, coke, charcoal and woodchips. The raw materials are fed into a furnace, where the raw material charge is penetrated by electrodes, in most cases three electrodes carrying three phase electric current, which passes between them and cancels in a star connection in the furnace. The current flows through the intermediate reaction products filling the furnace, and in many cases passes through an electric arc as it leaves the electrode. Ideally the stoichiometry of the over-all reaction in the furnace can be written as (Schei 1998):



This occurs through a series of sub-reactions at different locations in the furnace, depending on the local temperature for thermodynamic conditions to enable spontaneous reactions. The carbon materials react in the upper regions of the furnace with SiO gas formed further down

in the furnace to form SiC (Eq. 2) at temperatures above 1500°C. SiO gas also condenses to form a condensate made of a mixture of SiO₂ and Si (Eq. 3) in the colder regions of the furnace.



As the charge in the furnace is consumed from below, the SiC reaches the hotter, lower regions of the furnace where it finally reaches a high enough temperature to enable the Si forming reaction by reacting with SiO gas as shown in reaction 4, which requires a minimum of 1811°C. The SiO is formed by the reversed reaction 3, here denoted by 3*



The driving forces for both reactions 4 and 3* increase with increasing temperature, but the speed of reaction 4 increases faster than that of reaction 3*. Therefore the rate of Si production rises faster than the SiO production with increasing temperature, improving the Si yield. For good Si yield, these reactions should occur at almost 2000°C, and therefore the electric arcs, which generate heat at a very high temperature, are important for the process.

The CO gas formed in the furnace reacts with oxygen in the air at the top of the raw material charge in the furnace to form CO₂. Also, a part of the SiO gas formed by reaction 3* will not be captured and participate in reactions 4, 2 or 3, but instead it will escape and burn alongside the CO gas and form amorphous SiO₂ fumes that are captured in bag-house filters and sold into a number of applications.

It is the stoichiometry in equation 1, which accounts for most of the greenhouse gas emissions from the MG-silicon production process, decided by the four valent Si in quartz, and that the thermodynamics favor the formation of CO gas rather than CO₂. According to the stoichiometry around 3.26 kg of CO₂ should be formed for each kg of Si metal produced. This is unavoidable for a carbothermic process, as the carbon is needed to capture the oxygen from the quartz, leaving the metal behind. In reality, there are inefficiencies and losses in the process that increase the carbon consumption per unit metal produced. For example, volatiles in the raw materials evaporate and burn, releasing CO₂, without contributing to the reduction. Another example of such inefficiencies is that carbon is consumed in the production of SiO gas that escapes the process to form silica fumes reducing the fraction of incoming Si in quartz, which is tapped as metal, also termed *silicon yield*. Thus, the greenhouse gas emissions, as well as the energy consumption of the process, are closely linked to the silicon yield for the process. The amount of carbon consumed to produce silica fumes rather than metal is quantified with regard to the stoichiometric carbon consumption. The term stoichiometric carbon consumption normally refers to dry carbon used per Si in quartz going into the process so that 100% stoichiometric carbon consumption would correspond to the stoichiometry in equation 1. In most cases the carbon content of a raw material mix as per this definition will be below 100%, as some of the incoming silicon will leave the process as SiO, which forms silica fumes. In some of the CO₂ related literature the term is defined as per metal produced, which gives a percentage higher than 100%. To adhere to that convention while avoiding confusion the term stoichiometric carbon consumption_{metal} will be used, so stoichiometric carbon consumption_{metal} = 110% means you need 10% more carbon to produce your metal than given by the stoichiometry from equation 1 on the basis of tapped metal. In addition there are emissions from the electrode consumption. An estimate of the composition of a typical emission factor for MG-silicon production assuming 109% stoichiometric carbon consumption_{metal} and that 10% of the dry carbon content comes from woodchips published by (Monsen 1998) is shown in Table 1. The IPCC estimate is in agreement with Kero et al. (IPCC2006, Kero 2016) with

process emissions of 5 kg CO₂/kg Si, but less details on assumption are given, in particular it may be deduced from the IPCC report that the effect of woodchips is not considered in that publication. It is very difficult to operate a silicon furnace without woodchips, so therefore the process footprint from (Monsen 1998) is used for the estimates in this paper. Using the emission factor of 5 kg CO₂/kg Si would shift all numbers up by 0.32 kg CO₂/kg Si but does not change the analysis. As the focus of this paper is on the effect of the energy mix, this published carbon footprint for 109% stoichiometric carbon consumption_{metal} and that 10% of the carbon comes from biocarbon is used for all regions. Regional variation in the process carbon footprint based on use of biocarbon is a topic for another study.

Table I: Typical emissions for MG-silicon production based on estimate in (Monsen 1998), assuming 109% stoichiometric carbon consumption_{metal} and 10% of carbon coming from woodchips on dry basis. In this estimate, no use of charcoal is assumed.

	Consumption on dry basis, kg/tonne metal	Emission factor CO ₂	CO ₂ emission tonne/tonne Si metal
Coal	1390	3.12	+4.33
Electrode	100	3.54	+0.35
Emission Factor			=4.68

Reducing greenhouse gas emissions from the carbothermic process

In the carbothermic MG-silicon production process, woodchips are a part of the raw materials. The presence of woodchips improves the furnace operation as it improves the permeability of the charge, improving the capture of SiO gas and thus the silicon yield. The amount of woodchips used is very location dependent due to differing availability of this resource. Also charcoal made from biocarbon can with benefit substitute coal and coke in the process. The carbon provided by woodchips in the raw material mix is biocarbon and does not contribute directly to the greenhouse gas emissions. Also, most silicon smelters have charcoal as a part of their raw material mix in an effort to reduce the carbon footprint, not hurt by the fact that the high reactivity of charcoal facilitates the operation of the submerged arc furnace. The handling of biocarbon and production of charcoal will be associated with emissions, which can be estimated through life cycle analysis, but it is clear that replacing fossil carbon with woodchips and charcoal is a significant opportunity to reduce the carbon footprint (Riva 2019). However, the availability and cost of biocarbon are very location dependent. The carbon content by weight is much lower for woodchips than coal and coke, which increases the contribution of transportation in the raw material cost, and in most locations, charcoal will be more costly than fossil carbon sources. It can be economically feasible to run the furnace almost exclusively on locally sourced biocarbon in South America, for example Elkem's plant in Paraguay is running on 100% biocarbon, while most biocarbon would need to be imported to a production facility located in Iceland.

Carbon Capture and Storage (CCS)

Carbon capture and storage is capital intensive and would require incentives to become economically viable for metallurgical industries. Research and development have focused most efforts on processes that are larger in volume than ferroalloys, such as steel and cement production, but solution developed for those industries are likely to be transferrable to ferroalloy producers, such as silicon smelters (Wyns 2020). The concentration of CO₂ in the flue gas from the carbothermic process can be around 4%, which makes it compatible with existing technologies for carbon capture and up-concentration for either sequestering or reuse. The economy of such solutions would also be improved if infrastructure for carbon capture and transport is developed for larger industrial producers located close by.

New carbon capture-ready iron ore smelting technologies such as HIsarna, currently piloted at Tata Steel in IJmuiden in the Netherlands, show that CCS in theory can be used in non-ferrous metals smelting processes, in particular for carbothermic processes and it should therefore be applicable for the silicon process. However, the capture and cleaning of the dirty process gases before up-concentration will be technically challenging and costly, so the investment costs might be high compared to the total amount of emissions captured. If CCS is applied in combination with biocarbon such as woodchips and charcoal as large part of the carbon raw materials, the production can even be considered as carbon negative. There are several initiatives on how to sequester the captured CO₂, such as storing the gas in depleted oil reservoirs (Minardi 2021), or the permanent sequestering the CO₂ in the form of under ground minerals, as is the case in the Carbfix process (Clarc 2020).

Alternative production processes

There are initiatives to develop alternative processes to produce silicon by using alternative reduction agents for the quartz. A process based on the use of methane gas as a reduction agent in a plasma furnace has been proposed, which would significantly reduce CO₂ emissions but not eliminate them (Sörvik 2006, Aarnæs 2020). Hydrogen has been proposed as a promising reduction agent for iron smelting, but does not seem to offer a thermodynamically favorable chemistry to produce silicon metal, although it readily produces the intermediate SiO gas by reaction with SiO₂ (Han 2005). In the carbothermic process the final metal producing step is obtained by a reaction between the intermediate compounds SiO and SiC as shown in reaction 4, but an alternative silicon forming reaction seems elusive when using only hydrogen as a reduction agent. The reduction of SiO to Si by H₂ is not favored by thermodynamics. This obstacle can be overcome using methane as a reduction agent, as the carbon in methane will react with the silica to form SiC (Monsen 2013), enabling the metal forming reaction to occur. Condensation of SiO into a condensate containing SiO₂ and Si metal droplets forms metal, but a process based on this is likely to be very inefficient.

Electrochemical reduction in a molten salt, or directly from molten oxides, has also been proposed (Ge 2019, Su-Chul 2008), but is still in the laboratory scale and at a relatively low technology readiness level at this time.

ELECTRICAL POWER SOURCES FOR MG-SILICON PRODUCTION – THE INDIRECT CO₂e EMISSIONS

Historically, the deciding factor for the location of silicon and ferrosilicon plants was to be close to hydro-powered electrical energy because of the reliability of the supply, low transmission costs, and low-cost renewable energy. A number of plants were also historically built in regions with nuclear energy as a significant part of their energy mix. As silicon production is very energy intensive, one of the highest potential routes to reduce emissions from the production is to reduce the indirect emissions due to the production of electric power used in the carbothermic process.

The data in Table II, based on data from CRU, includes electric power sources used for MG-silicon production in submerged arc furnaces, process, including auxiliary energy used to drive fans, etc.

In this period, from 1995 to 2019, the production of silicon has increased by 456%, the increase being 239% since 2000. Most of this capacity increase has occurred in China. More silicon is produced by hydropower and other renewables than ever before and the proportion produced

with renewables has only declined from 55% to 53% in this period. This is in sharp contrast to aluminum production, where the portion of renewables in the energy mix has declined from 71% to 27% from 2000 to 2018, while production has, similarly to the silicon production, increased by 261% (Saevarsdottir 2020a).

Table II: Electrical energy numbers and power sources in TWh/year for MG-silicon production per region for the years 1995, 2000, 2010 and 2019 along with the production per region and change in production over the period. Production and specific energy numbers are based on data from CRU*, while other energy numbers are based on country mixes.

	EEA	NA	Russia	China	South America	Other	Total
Ren. (2019)	3.01	0.99	0.31	12.55	2.08	0.24	19.17
Ren. (2010)	2.28	1.16	0.32	8.04	2.18	0.19	14.16
Ren. (2000)	2.29	1.37	0.42	2.57	1.89	0.04	8.59
Ren. (1995)	1.52	1.13	0.00	0.00	1.57	0.04	4.26
Nuclear (2019)	0.99	0.30	0.00	0.00	0.08	0.01	1.38
Nuclear (2010)	1.18	0.37	0.00	0.00	0.08	0.03	1.66
Nuclear (2000)	1.02	0.42	0.00	0.00	0.04	0.03	1.51
Nuclear (1995)	0.83	0.38	0.00	0.00	0.02	0.02	1.26
Coal (2019)	0.16	0.36	0.31	12.77	0.11	0.78	14.50
Coal (2010)	0.28	0.50	0.32	6.49	0.05	0.95	8.59
Coal (2000)	0.44	0.56	0.42	2.02	0.07	0.76	4.27
Coal (1995)	0.38	0.45	0.00	0.00	0.03	0.67	1.54
NG (2019)	0.21	0.45	0.00	0.00	0.26	0.27	1.19
NG (2010)	0.21	0.53	0.00	0.00	0.19	0.10	1.03
NG (2000)	0.14	0.59	0.00	0.00	0.02	0.04	0.78
NG (1995)	0.13	0.51	0.00	0.00	0.00	0.03	0.67
Other (2019)	0.01	0.02	0.00	0.00	0.00	0.00	0.02
Other (2010)	0.01	0.01	0.00	0.00	0.00	0.00	0.03
Other (2000)	0.01	0.01	0.00	0.00	0.00	0.00	0.03
Other (1995)	0.01	0.01	0.00	0.00	0.00	0.00	0.01
Total Si prod. in 2019 (t/year)	362.50	165.00	48.20	1895.00	202.00	106.50	2779.20
Total Si prod. in 2010 (t/year)	326.00	197.00	48.74	1115.00	204.50	102.50	1993.74
Total Si prod. in 2000 (t/year)	316.00	225.00	65.00	352.50	168.50	38.00	1165.00
Total Si prod. in 1995 (t/year)	231.50	184.50	0.00	0.00	133.00	60.00	609.00
Change from 1995 to 2019	157%	89%	-	-	152%	178%	456%

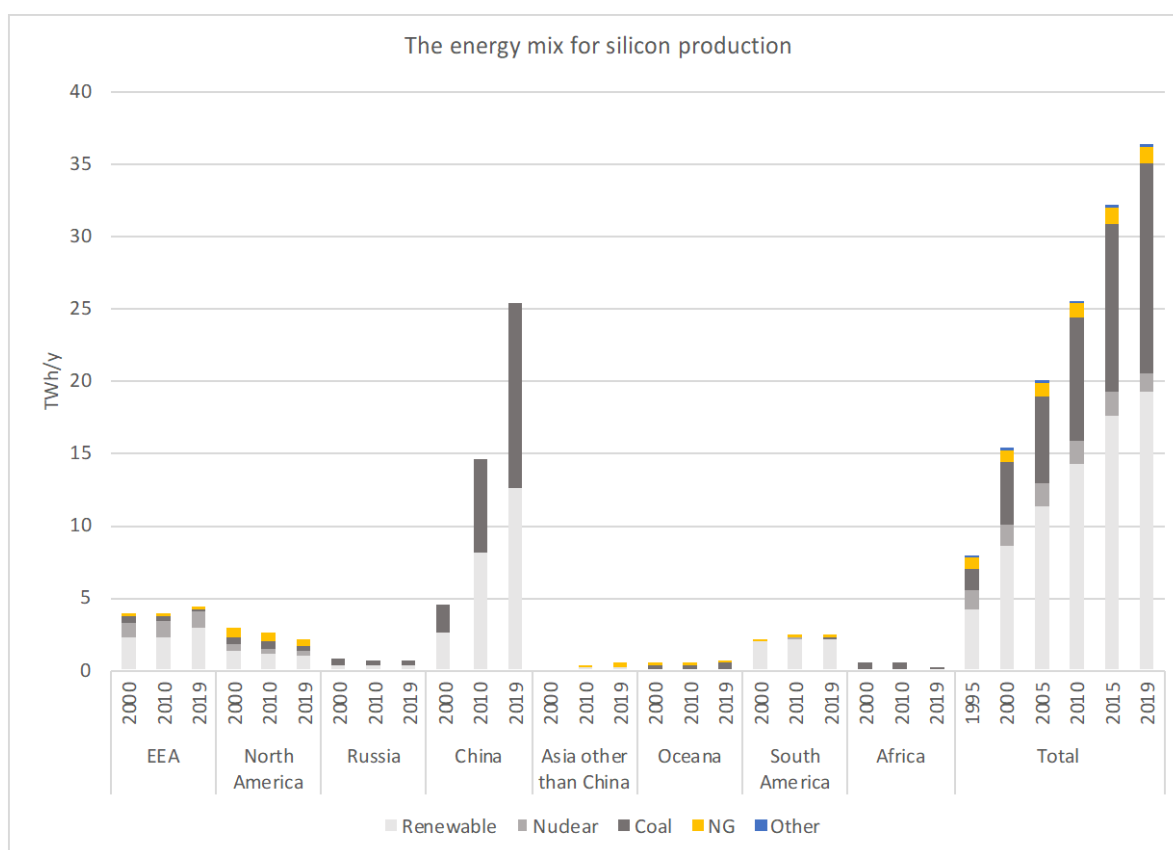


Figure 1: Energy mix for MG-silicon production (TWh/y) by region for the years 2000, 2010 and 2019, along with the total energy mix for 1995 to 2019 based on information from CRU.

An important reason for this is that silicon smelters are relatively small units as compared to aluminum smelters, for example, and the furnaces can be shut down without permanent damage to equipment. This has enabled new silicon production capacity in China being to a larger extent built close to hydropower sources, adapting their power use for supply. In fact, in China about 5 times more silicon is currently produced using hydro power as compared to 2000, and this is close to half of their total production.

However, in this period since 1995, less is produced with nuclear energy, the contribution to the silicon power mix declining from 14% to 4%. Due to this the fraction of the silicon energy mix coming from low-carbon energy sources has declined from 71% to 57%.

The rest of the energy mix is mostly supplied by coal thermal power, which has doubled its share from 20% to 40% in the period, and which has great implications for the average carbon footprint for this production.

Europe excluding CIS is denoted by the European Economic Area (EEA) in Table II to account for Norway and Iceland in addition to the EU, has the highest proportion of low carbon energy in the silicon mix, or close to 90%, thereof 20% being nuclear energy in France. Most of the renewable contributions being hydropower in Norway and Iceland, but Germany and Spain still have some production, partly powered by renewables. Asia other than China, Oceania and Africa are grouped together in Table II due to data sensitivity.

Figure 1 is a graphical representation of some of the data in Table II and shows the same trends. It is seen how Europe has slightly increased their silicon metal production and North America has decreased its production, while China has increased its silicon production tremendously.

Figure 2 shows how the carbon footprint of MG-silicon production in kg CO₂e /kg Si, including both the emission factor from the production process (Monsen 1998) and the indirect emissions from the electrical energy used for silicon production. The figure shows how the carbon footprint for the energy mix per region has changed with time, as well as the global average. Variability of the process carbon footprint by region is not shown here, but is likely to be significant due to different share of biocarbon used in the raw material mix.

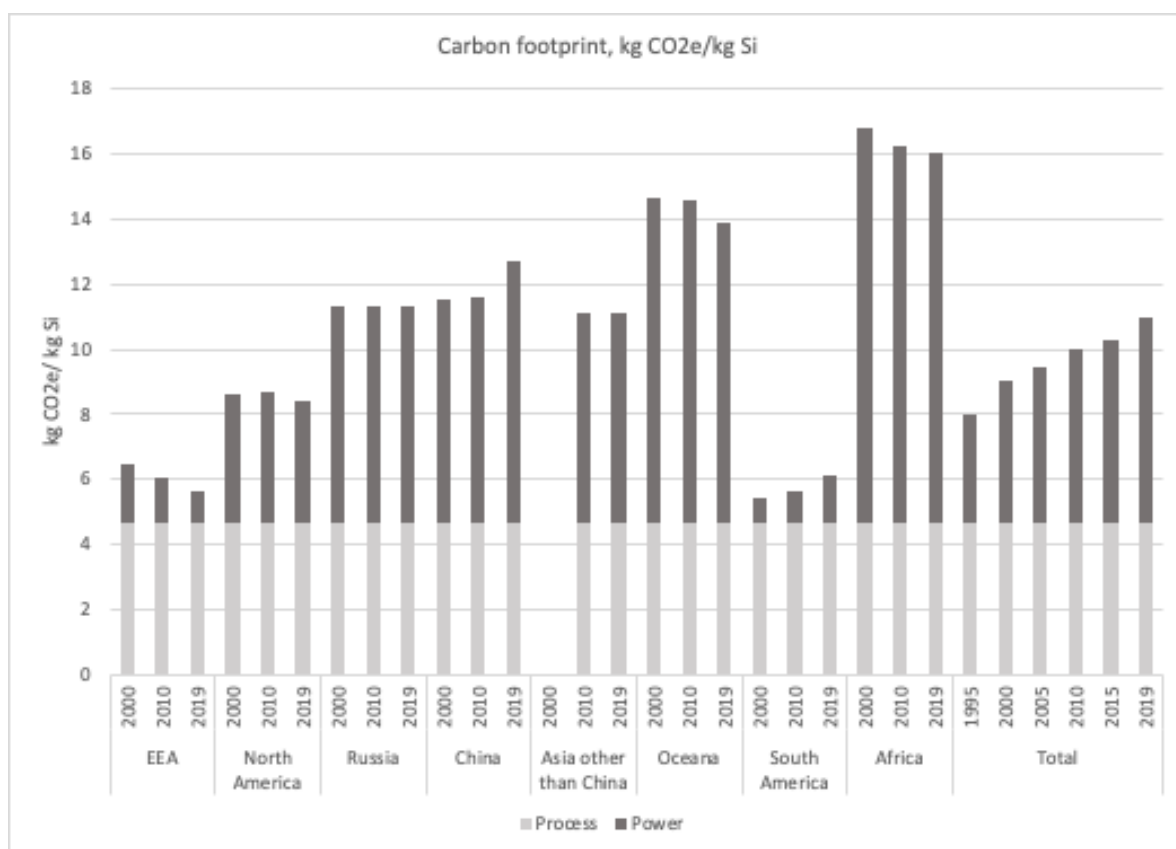


Figure 2: The overall carbon footprint of MG-silicon production (kg CO₂e/kg Si) based on average emission values attributed to the electrical energy used for silicon production by region in the years 2000, 2010 and 2019, and the energy emission values published by IPCC (Bruckner 2014). Also the average global emissions for this production are shown for the years 1995 to 2019. Regional variations in the process emissions due to the share of biocarbon in the reductant mix are not accounted for in this work. The emission factor for the direct process emissions is kept constant, assuming 10% of dry carbon content comes from woodchips (Monsen 1998).

From Figures 1 and 2 it can be seen that the carbon footprint, here described in terms of intensity per kg Si, (kg CO₂e/kg Si) from the electrical energy used in silicon production, has decreased significantly in Europe as the energy production has on average shifted mostly to green power. Also the specific emissions have decreased slightly in other regions due to process improvements leading to lower specific energy consumption.

The reduced specific energy consumption is likely associated with an improved silicon yield, which decreases emissions per tonne produced, but the published information of emission factor from the process does not offer that detail, so such speculations will not be indulged in this paper. The reason for the observed increase in the carbon footprint in the power related emissions is solely due to a shift in production towards China, which currently has a 68% share of the global MG-silicon production, up from 30% in 2000, and very little in 1995. The indirect carbon footprint was 3.3 kg CO₂e/kg Si in 1995, but had risen to 6.3 kg CO₂e/kg Si, or by 90%. The increase from the year 2000 when it was 4.4 kg CO₂e/kg Si, is 43%. As the process emission

factor is assumed constant at 4.7 kg CO₂e/kg Si, this equals an increase in the total carbon footprint from 8 to 11 kg CO₂e/kg Si, and currently around 57% of the carbon footprint is coming from the electric power production. This can be compared to the development of the carbon footprint for aluminum production, which has increased by 3 kg CO₂e/kg Al since 2000, as a result of an increase in production in China based 90% on coal thermal power (Saevarsdottir 2020b).

CARBON LEAKAGE

It is clear that hydropower and other renewables to a smaller extent are very important to reduce the global carbon footprint for MG-silicon production. If all the production were based on coal power, the energy related emissions would be 11 kg CO₂e/kg Si, bringing the overall carbon footprint up to 15.7 kg CO₂e/kg Si, in which case the energy related emissions would contribute 70% to the global carbon footprint rather than 57% as is currently the case.

The strong historical and still significant contribution of nuclear energy in reducing the carbon footprint of primary production of metals should be noted. As close to a third of the European MG-silicon production is located in France, mostly based on nuclear energy, this energy source contributes to reducing the average global carbon footprint from the energy used in this process by 0.42 kg CO₂e/kg Si or by 8.7% as compared to if that power was produced by coal thermal power plants. In 1995 nuclear power reduced the average global carbon footprint by around 1.8 kg CO₂e/kg Si or by 60% as compared to using coal instead of nuclear. It seems clear that the push to close nuclear power plants runs counter to the goal of reduced climate gas emissions, while renewable energy to fill the gap is still in short supply.

In general plants operated in regions that offer access to renewable energy or nuclear, such as Norway, Iceland, France and South America play a significant role in limiting the global emissions from metals that would otherwise be produced with electrical energy from fossil fuels. However, this silicon production has the potential to introduce very high *carbon leakage*.

Carbon leakage occurs when there is an increase in carbon dioxide emissions in one country (or region) as a result of emissions reduction by a second country with a strict climate policy. As production is initiated in the first country it replaces capacity closed down in the latter.

The European Union has developed a system for *carbon taxing*, the Emission Trading System (ETS), where industry must purchase quotas for their emissions. The EU is aware of the risk of carbon leakage and has identified businesses at a particular risk of carbon leakage that get awarded free emission quotas. Industries with high international trading intensity, and where emission costs are a significant fraction of revenue, are deemed at risk of carbon leakage, and ferroalloy production, including silicon, scores very high in both of those dimensions. Being very energy intensive, silicon production is seen as particularly sensitive to the effect of the ETS system on the energy price. The silicon industry is therefore also present on a much shorter list of businesses, which qualify for government subsidies to make compensate for the increase in electricity price as a consequence of the ETS system, as the most important threat to the competitiveness of the European silicon industry is the energy price.

Most countries have chosen to activate the authorization to subsidize the indirect emission energy cost to this sector, but the risk is that silicon smelters will close down in countries that will not subsidize, as recent smelter closures in Iceland show.

In a wider sense the expression of carbon leakage implies greenhouse gas emissions, global management and responsibility. Closing down hydro powered smelters to reduce the emissions regionally by eliminating the contribution from the carbothermic process will increase global emissions, because the deficit in metal supply will almost certainly be compensated by production in locations using a higher share of carbon intensive power sources.

CONCLUSIONS

This paper summarizes the development in the carbon footprint from MG-silicon production with focus on the indirect emissions from the electric power used in the process. It is shown that the carbon footprint of MG-silicon can range from 4.7 kg CO₂e/kg Si to 16 kg CO₂e/kg Si depending on the source of electric power used in the process, the lowest footprint being obtained if renewable or nuclear energy is used, while the coal thermal power leads to triple carbon footprint. The production of MG-silicon in 2019 has increased by a factor of 4.56 from 1995, and by a factor of 2.4 from 2000. Most of that increase has taken place in China, which contributes with 68% of the global production of this commodity. Even though almost half the Chinese production is with renewable energy, this has led to an increase in the total carbon footprint of MG-silicon from 8 to 11 kg CO₂e/kg Si from 1995 to 2019, and now the indirect emissions from the electric power production contribute around 57% of the overall carbon footprint.

This is a step in the wrong direction for a world striving for carbon neutrality, and it is clear that decarbonizing the energy sources used in silicon production, in addition to addressing the emissions from the process itself, is necessary to hamper climate change.

ACKNOWLEDGEMENTS

Much of the industrial production data used in this paper has been sourced from CRU, and were gracefully shared by Mr. Jorn de Linde. His and CRU's contributions are gratefully acknowledged.

REFERENCES

- Aarnæs 2020, Aarnæs, T.S., Ringdalen, E. and Tangstad, M. Silicon carbide formation from methane and silicon monoxide. *Sci Rep* **10**, 21831 (2020). <https://doi.org/10.1038/s41598-020-79006-6>
- Bruckner 2014, Bruckner T., Bashmakov I.A., Mulugetta Y., Chum H., de la Vega Navarro A., Edmonds J., Faaij A., Fungtammasan B., Garg A., Hertwich E., Honnery D., Infield D., Kainuma M., Khennas S., Kim S., Nimir H.B., Riahi K., Strachan N., Wiser R. and Zhang X. (2014) Energy Systems. In: Climate Change 2014: Mitigation of Climate Change. Contribution of Working Group III to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change. Edenhofer O., Pichs-Madruga R., Sokona Y., Farahani E., Kadner S., Seyboth K., Adler A., Baum I., Brunner S., Eickemeier P., Kriemann B., Savolainen J., Schlömer S., von Stechow C., Zwickel T. and Minx J.C. (eds.)). Cambridge University Press, Cambridge, United Kingdom and New York, NY, USA. https://www.ipcc.ch/site/assets/uploads/2018/02/ipcc_wg3_ar5_full.pdf. Accessed 4 October 2019.
- Clark 2020, Clark D.E., Oelkers E.H., Gunnarsson I., Sigfússon B., Snæbjörnsdóttir S. Ó., Aradóttir E. A., and Gíslason S. R. (2020) [CarbFix2: CO₂ and H₂S mineralization during 3.5 years of continuous injection into basaltic rocks at more than 250 °C](#) *Geochimica et Cosmochimica Acta*, Vol 279, 45-66.
- Ge 2019, Ge J., Zou X., Almassi S., Ji L., Chaplin B.P. and Bard A. J. *Angew. Chem. Int. Ed.* **2019**, 58, 16223.
- Han 2005, Han G. and Sohn H. Y. Kinetics of the Hydrogen Reduction of Silica Incorporating the Effect of Gas-Volume Change upon Reaction, *J. Am. Ceram. Soc.*, 88 [4] 882-888 (2005) DOI: 10.1111/j.1551-2916.2005.00144.x

- Kero 2016, Kero I, Gradahl S and Tranell G. Emissions from MG-Si/FeSi production – a summary, Silicon for the Chemical and Solar Industry XIII, Kristiansand, Norway, June 13-16, 2016.
- IPCC 2006, Faerden T, Lubetsky J.S, Lindstad T, Olsen S.E and Tranell G. 2006 IPCC Guidelines for National Greenhouse Gas Inventories.
- IPCC 2018, Summary for Policymakers. In: Global Warming of 1.5°C, an IPCC special report on the impacts of global warming of 1.5°C above pre-industrial levels and related global greenhouse gas emission pathways, in the context of strengthening the global response to the threat of climate change, sustainable development, and efforts to eradicate poverty. Masson-Delmotte V., Zhai P., Pörtner H.-O., Roberts D., Skea J., Shukla P.R., Pirani A., Moufouma-Okia W., Péan C., Pidcock R., Connors S., Matthews J.B.R., Chen Y., Zhou X., Gomis M.I., Lonnoy E., Maycock T., Tignor M. and Waterfield T. (eds.)). *World Meteorological Organization, Geneva, Switzerland*, 32 pp.
- Monsen 2013, Monsen B., Kolbeinsen L., Prytz S., Myrvagnes V. and Tang K., Possible Use of Natural Gas For Silicon or Ferrosilicon Production, The 13th International Ferroalloys Congress (Infacon13) Almaty, Kazakhstan, June 2013.
- Monsen 1998, Monsen, B., Lindstad T. and Tuset J. (1998). CO₂ Emissions from the Production of Ferrosilicon and Silicon metal in Norway.
- Minardi 2021, Minardi A., Stavropoulou E., Kim T., Ferrari A. and Laloui L., Experimental assessment of the hydro-mechanical behaviour of a shale caprock during CO₂ injection, *International Journal of Greenhouse Gas Control*, Volume 106, 2021, 103225, ISSN 1750-5836.
- Riva 2019, Riva L., Nielsen. H. K., Buø T.V., Zhuo H., Yang Q., Yang H., Skreiberg Q., Wang L., Bartocci P., Barbanera M., Fantozzi F. „Analysis of Biocarbon Pellet Production to Substitute Coke“, Joint International Conference on Energy, Ecology and Environment and Electrical Intelligent Vehicles (ICEEE 2019/ICEIV 2019) ISBN: 978-1-60595-641-1 LCA.
https://www.researchgate.net/publication/336931020_LCA_Analysis_of_Biocarbon_Pellet_Production_to_Substitute_Coke [accessed Jan 22 2021].
- Saevarsdottir 2020a, Saevarsdottir G., Kvande H. and Welch B.J. (2020) Reducing the Carbon Footprint: Aluminium Smelting with Changing Energy Systems and the Risk of Carbon Leakage. In: Tomsett A. (eds) *Light Metals 2020. The Minerals, Metals & Materials Series*. Springer, Cham. https://doi.org/10.1007/978-3-030-36408-3_98.
- Saevarsdottir 2020b, Saevarsdottir, G., Kvande, H. & Welch, B.J. Aluminum Production in the Times of Climate Change: The Global Challenge to Reduce the Carbon Footprint and Prevent Carbon Leakage. *JOM* **72**, 296–308 (2020). <https://doi.org/10.1007/s11837-019-03918-6>
- Schei 1998, Schei A., Tuset J.K. and Tveit H. Production of High Silicon Alloys, Tapir Forlag, Trondheim 1998.
- Su-Chul 2008, Su-Chul L., Jin-Mok H. and Chung-Seok S. Silicon powder production by electrochemical reduction of SiO₂ in molten LiCl–Li₂O, *Journal of Industrial and Engineering Chemistry*, Volume 14, Issue 5, 2008, Pages 651–654, ISSN 1226-086X.
- Sörvik 2006, Sörvik A. I. International patent no: PCT/NO2007/000092.
- World Aluminum 2017, “World Aluminium Publications, “A life-cycle model of Chinese grid power and its application to the life cycle impact assessment of primary aluminium”, <http://www.world-aluminium.org/publications/tagged/life%20cycle/>. Accessed September 9, 2019.
- Wyns 2020, Wyns T. and Khandekar G. Metals for a Climate Neutral Europe A 2050 Blueprint, The Institute for European Studies, Vrije Universiteit Brussel (VUB).
https://www.ies.be/files/Metals_for_a_Climate_Neutral_Europe.pdf accessed Jan. 6th 2021.



Gudrun Saevarsdottir

Associate professor, Reykjavik University

I hold BS and MS degrees in physics from the University of Iceland and a PhD in Materials Science and Engineering from NTNU. I am currently an associate professor in Engineering at Reykjavik University and my research interests pertain to reducing greenhouse gas emissions from industrial processes. I am currently involved in research on silicon/ferrosilicon production, aluminum production and geothermal energy, all leveraging fundamentals in thermodynamics, fluids and electromagnetism to improve efficiency and reduce emissions.